SNAPLOGIC INFRASTRUCTURE ANALYSIS • CONFIDENTIAL • PROFESSIONAL SERVICES INPUT

Syngenta – Infrastructure Analysis & Target Architecture

Current-state SnapPlex inventory, workload analysis, PS pipeline candidates, CI/CD parity gaps & node sizing recommendation

Data as of 26 March 2026 | Source: BigQuery codedugong | 90-day execution window | 3 orgs · 7 Snaplexes · 19 active nodes

PROD01 AVG CPU
~68%
⚠ Down from 81% (Mar 3) — still high

NODES TO ADD
+6
+12 vCPUs for DnA dev & test CI/CD parity

DEV FAILURE RATE
86%
⚠ 603K nested pipeline failures (90d)

PROD02 (DNA)
New
Will offload workloads from prod01 as DnA ramps

CI/CD GAPS
2
DnA plex missing in Test & Dev

🔴 **Prod01 still at high CPU utilisation:** Despite dropping from 81% (March 3) to ~68% average over the past 30 days, SyngentaGroundplex prod01 nodes regularly peak at 100%. Nested pipelines averaging 111 hours of runtime are continuously saturating nodes. Additional compute is still required.

🟣 **DnAGroundplex (prod02) will offload prod01:** The new 4-node Groundplex at `SyngentaProd/rt/sidekick/prod02` is active and currently at ~2% CPU as DnA workloads onboard. As pipelines migrate to prod02, this directly reduces execution pressure on the main prod01 Groundplex — helping bring its average CPU below the 60% target. **No equivalent DnA plex exists yet in Test or Dev — this is the primary CI/CD gap to address.**

Provisioning: Current State vs Target — All Snaplexes

What is deployed today vs what is recommended based on workload and CI/CD requirements. Node sizing assumes same physical spec as existing on-prem nodes (4 vCPU / 15.8 GB RAM for prod, 2 vCPU / 7.9 GB RAM for dev/test) unless noted.

ENVIRONMENT	SNAPLEX / PATH	TYPE	CURRENT NODES	VCPUS NOW	AVG CPU	RECOMMENDED	TARGET VCPUS	Δ VCPUS	RATIONALE
Prod	`SyngentaProd/rt/sidekick/prod01`	Groundplex · On-prem	9 nodes	36	**68% avg**	**9 nodes (keep)**	36	0	No additional nodes required. prod02 (DnA) will offload workloads as DnA ramps, reducing prod01 pressure. Monitor CPU trajectory over next 60 days.
Prod	`SyngentaProd/rt/sidekick/prod02`	Groundplex · On-prem (DnA)	4 nodes	16	2% avg	**4 nodes (keep)**	16	0	Monitor 60 days. If DnA workload scales, add 2 more. Not yet justified — do not over-provision.
Test	`SyngentaTest/rt/sidekick/test01`	Groundplex · On-prem	3 nodes (1 idle)	6	~15% avg	**5 nodes (+2)**	10	**+4**	Reclaim idle node first. Add 2 active nodes to support CI/CD validation of prod-scale concurrent patterns. Node spec: 2 vCPU / 7.9 GB.
Test	`SyngentaTest/rt/sidekick/dna-test01` (new)	Groundplex · On-prem (DnA mirror)	0 — does not exist	—	CI/CD gap	**+2 nodes (new plex)**	4	**+4**	Required for DnA pipeline CI/CD gating before prod promotion. Normal nodes (2 vCPU / 7.9 GB) — consistent with test01 spec. Minimum 1 JCC + 1 worker.
Dev	`syngenta_dev/rt/sidekick/dev01`	Groundplex · On-prem	4 nodes	8	~28% avg	**4 nodes (keep)**	8	0	Sufficient compute. Root issue is 86% pipeline failure rate — hardware won't fix that. PS review needed first.
Dev	`syngenta_dev/rt/sidekick/dna-dev01` (new)	Groundplex · On-prem (DnA mirror)	0 — does not exist	—	CI/CD gap	**+2 nodes (new plex)**	4	**+4**	Mirror of test DnA plex for developer iteration. Normal nodes (2 vCPU / 7.9 GB) — consistent with dev01 spec.
Dev	`syngenta_dev/rt/cloud/dev`	**Ultraplex (SnapLogic-hosted)**	2 JCC + 2 FM	12	1–3% avg	**Trial end**	—	—	Trial instance — not relevant for production licensing. End of trial, no further action required.
Dev	`syngenta_dev/rt/cloud/autosync`	Cloud Snaplex · SnapLogic-hosted	1 JCC node	2	1.4% avg	**Trial end**	—	—	Trial instance — not relevant for production licensing. End of trial, no further action required.
TOTALS							**78 vCPUs**	**+12 vCPUs**	**DnA test & dev mirrors (+8) + test groundplex expansion (+4)**

1 Current SnapPlex Inventory & Node Metrics (30-day)

All nodes with current configuration and recent CPU/memory utilisation. Node utilisation sourced from node_usage_us (data through Mar 4) and snaplex_us.telemetry_hour (current).

Production **SyngentaProd** 13 nodes · 2 Groundplexes · 52 vCPUs · ~206 GB RAM

SYNGENTAGROUNDPLEX | SYNGENTAPROD/RT/SIDEKICK/PROD01 | GROUNDPLEX · ON-PREMISES 9 NODES 36 VCPUS

NODE HOSTNAME	VCPU	RAM	JVM HEAP	MAX SLOTS	CPU AVG (30D)	CPU MAX (30D)	MEM AVG
deawiwsplp0009	4	15.8 GB	13.4 GB	4,000	74%	100%	53%
deawiwsplp0003	4	15.8 GB	13.4 GB	4,000	74%	100%	46%
deawiwsplp0013	4	15.8 GB	13.4 GB	4,000	73%	100%	46%
deawiwsplp0001	4	15.8 GB	13.4 GB	4,000	72%	100%	51%
deawiwsplp0011	4	15.8 GB	13.4 GB	4,000	69%	100%	55%
deawiwsplp0002	4	15.8 GB	13.4 GB	4,000	67%	100%	55%
deawiwsplp0012	4	15.8 GB	13.4 GB	4,000	65%	100%	51%
deawiwsplp0004	4	15.8 GB	13.4 GB	4,000	59%	100%	52%
deawiwsplp0010	4	15.8 GB	13.4 GB	4,000	55%	100%	47%
TOTALS / AVERAGES				**36,000**	**Avg 68%**	**Max 100%**	**Avg 51%**

DNAGROUNDPLEX | SYNGENTAPROD/RT/SIDEKICK/PROD02 | GROUNDPLEX · ON-PREMISES NEW SINCE MAR 2026 4 NODES 16 VCPUS

NODE HOSTNAME	VCPU	RAM	JVM HEAP	MAX SLOTS	CPU AVG (30D)	CPU MAX (30D)	MEM AVG
deawilsplp0003	4	15.4 GB	14.0 GB	4,000	2%	98%	~5%
deawilsplp0001	4	15.4 GB	14.0 GB	4,000	2%	100%	~5%
deawilsplp0002	4	15.4 GB	14.0 GB	4,000	2%	100%	~5%
deawilsplp0004	4	15.4 GB	14.0 GB	4,000	2%	100%	~5%
TOTALS / AVERAGES				**16,000**	**Avg 2%**	**Max 100% (spike)**	**~5%**

Test **SyngentaTest** 3 nodes · 1 Groundplex · 6 vCPUs · 23.7 GB RAM

SYNGENTAGROUNDPLEX | SYNGENTATEST/RT/SIDEKICK/TEST01 | GROUNDPLEX · ON-PREMISES

NODE HOSTNAME	VCPU	RAM	JVM HEAP	MAX SLOTS	CPU AVG (30D)	CPU MAX (30D)	MEM AVG
deawiwsplt0001	2	7.9 GB	6.1 GB	4,000	22%	100%	18%
deawiwsplt0002	2	7.9 GB	6.3 GB	4,000	21%	100%	18%
deawiwsplt0003	2	7.9 GB	6.3 GB	4,000	3%	100%	2%
TOTALS / AVERAGES				**12,000**	**Avg 15%**	**Max 100%**	**Avg 13%**

Dev **syngenta_dev** 7 active nodes · 3 Snaplexes · 18 vCPUs · ~69 GB RAM

SYNGENTAGROUNDPLEX | SYNGENTA_DEV/RT/SIDEKICK/DEV01 | GROUNDPLEX · ON-PREMISES

NODE HOSTNAME	VCPU	RAM	JVM HEAP	MAX SLOTS	CPU AVG (30D)	CPU MAX (30D)	MEM AVG
deawiwspld0002	2	7.9 GB	6.1 GB	4,000	32%	100%	24%
deawiwspld0004	2	7.9 GB	6.3 GB	4,000	30%	100%	26%
deawiwspld0003	2	7.9 GB	6.3 GB	6,000	25%	100%	22%
deawiwspld0001	2	7.9 GB	6.1 GB	4,000	25%	100%	5%
TOTALS / AVERAGES				**18,000**	**Avg 28%**	**Max 100%**	**Avg 19%**

CLOUD (DEV) | SYNGENTA_DEV/RT/CLOUD/DEV | CLOUD SNAPLEX · SNAPLOGIC-HOSTED | TRIAL END

NODE IDENTIFIER	TYPE	VCPU	RAM	MAX SLOTS	CPU AVG (30D)	CPU MAX
pa22sl-jcc-ix0cf1…	JCC	4	15.3 GB	2,000	2%	88%
pa22sl-jcc-ix07f6…	JCC	4	15.3 GB	2,000	1%	88%
pa22sl-fm-ix0aad…	FM	2	7.6 GB	2,000	1%	88%
pa22sl-fm-ix033a…	FM	2	7.6 GB	2,000	3%	88%

AUTOSYNC | SYNGENTA_DEV/RT/CLOUD/AUTOSYNC | CLOUD SNAPLEX · SNAPLOGIC-HOSTED | TRIAL END

NODE IDENTIFIER	TYPE	VCPU	RAM	CPU AVG (30D)	LAST PIPELINE RUN
pa22sl-jcc-ux05db…	JCC	2	7.6 GB	1.4%	Feb 16, 2026 (nested_pipeline) · occasional manual runs only

2 CI/CD Parity — Prod vs Dev/Test

Every production Snaplex should have a mirror in dev and test to allow proper pipeline promotion and CI/CD validation.

🏭 Production (SyngentaProd)

SyngentaGroundplex PRESENT

path: prod01 · 9 nodes · 36 vCPUs

CPU: ~68% avg · All batch & integration

DnAGroundplex NEW

path: prod02 · 4 nodes · 16 vCPUs

CPU: ~2% avg · Data & Analytics workloads

🧪 Test (SyngentaTest)

SyngentaGroundplex PRESENT ⬆ RESIZE

path: test01 · **3 nodes today → target: 5 nodes · 10 vCPUs**

CPU: ~21% avg · 1 node idle · add 2 active nodes

▶ **Recommended: 5 nodes · 10 vCPUs**

DnA equivalent MISSING

No DnA Groundplex equivalent in Test

⚠ CI/CD gap for DnA pipelines

▶ Recommended: 2 nodes · 4 vCPUs

🛠 Dev (syngenta_dev)

SyngentaGroundplex PRESENT

path: dev01 · 4 nodes · 8 vCPUs

CPU: ~28% avg · active but high failure rate

DnA equivalent MISSING

No DnA Groundplex equivalent in Dev

⚠ CI/CD gap for DnA pipelines

▶ Recommended: 2 nodes · 4 vCPUs

⚠️ **CI/CD Gap — DnAGroundplex has no dev or test mirror.** The new `SyngentaProd/rt/sidekick/prod02` Groundplex (DnAGroundplex) has no equivalent in SyngentaTest or syngenta_dev. Any pipelines deployed on this plex cannot be properly validated in lower environments before production promotion. Syngenta should provision a DnA Groundplex equivalent in both dev and test as a priority action alongside onboarding DnA workloads.

⚠️ **Test Groundplex (test01): target 5 nodes — recommended sizing change.** Currently 3 nodes (6 vCPUs, 1 effectively idle) vs prod01's 9 nodes (36 vCPUs). Add 2 active nodes to reach 5 nodes / 10 vCPUs. This plex handles all non-DnA integration and batch pipeline CI/CD gating. Pipelines dependent on concurrent slot capacity or parallel execution cannot be properly validated at 3 nodes.

🔵 **Revalidation: 5-node target for test01 confirmed despite DnA plex addition and prod01 offload.**
Two factors considered: (1) *dna-test01 will be added* — DnA workloads move to their own dedicated test plex, keeping the workload separation clean. test01 remains the CI/CD environment for all non-DnA integration/batch pipelines. (2) *prod01 is offloading to prod02* — this reduces prod01's overall load but does not eliminate the need for test01 to validate prod-scale concurrent patterns for non-DnA workloads. Conclusion: 5 nodes on test01 is still the right target. The workload split between test01 (integration/batch) and dna-test01 (DnA) is consistent with the prod01/prod02 split — this is the correct CI/CD architecture.

3 Node Sizing — How Many Additional Nodes Are Needed

Capacity model based on current utilisation, target headroom (≤60% average CPU), and CI/CD parity requirements.

Sizing Model — SyngentaGroundplex (prod01)

PARAMETER	CURRENT	TARGET	NOTES
Node count	9	**9 (keep)**	No additional nodes required at this time
vCPUs	36	**36**	DnA workload migration to prod02 will reduce load
Current avg CPU	~68%	**≤60%**	Expected to improve as prod02 absorbs DnA workloads
prod02 impact	**Offload in progress**		4 new DnA nodes on prod02 will reduce prod01 execution pressure as DnA pipelines migrate

Recommendation: Hold at **9 nodes** on prod01. The 4 newly provisioned DnA nodes on prod02 will offload workloads from prod01 as DnA pipelines ramp. Monitor CPU trend over 60 days — if prod01 remains above 60% after DnA migration, reassess node additions then.

Sizing Model — DnAGroundplex (prod02) & Dev/Test Parity

ENVIRONMENT	PLEX	CURRENT NODES	RECOMMENDED	RATIONALE
Prod	DnAGroundplex	4 (2%)	**4 (keep)**	Monitor for 60 days. If DnA workloads ramp, add 2 more then. Do not remove — used for burst.
Test	DnAGroundplex (missing)	0	**+2 nodes**	Minimum viable test plex for DnA pipeline CI/CD gating. 2 nodes (1 JCC + 1 standby). Normal nodes (2 vCPU) — consistent with test01 approach.
Dev	DnAGroundplex (missing)	0	**+2 nodes**	Mirror of test plex for DnA developer workflows. Normal nodes (2 vCPU) — consistent with dev01 approach.
Test	SyngentaGroundplex	3 (incl. 1 idle)	**+2 nodes**	Investigate idle node first. If confirmed idle, reclaim and add 2 active nodes to support CI/CD validation of prod-scale pipeline patterns.

Total Additional Node Count

LOCATION / PLEX	NODES TO ADD	VCPUS ADDED	PRIORITY
SyngentaTest — DnAGroundplex (new plex)	**+2**	+4 vCPUs	HIGH
syngenta_dev — DnAGroundplex (new plex)	**+2**	+4 vCPUs	HIGH
SyngentaTest — SyngentaGroundplex (after idle node review)	**+2**	+4 vCPUs	MEDIUM
TOTAL ADDITIONAL NODES	**+6 nodes**	**+12 vCPUs**	

ℹ️ AutoSync and Ultra CloudPlex are trial instances and are not relevant for production licensing. Node counts above reflect only CI/CD-required infrastructure. prod01 node additions have been removed from scope — prod02 DnA (4 nodes provisioned) will handle the offload.

4 Licensing Summary — Target State

Target vCPU allocation split by environment tier, with Medium Node equivalent for licensing purposes. *Medium Node = 2 vCPUs (SnapLogic standard SKU)*. Trial instances (Cloud dev, AutoSync) excluded.

TIER	PLEXES INCLUDED	VCPUS	MEDIUM NODE EQUIV.
Production	prod01 (SyngentaGroundplex), prod02 (DnAGroundplex)	**52**	**26**
Dev / Test (Sandbox)	test01 (SyngentaTest), dna-test01 (DnATest), dev01 (SyngentaDev), dna-dev01 (DnADev)	**26**	**13**
Total (Licensed)	All active plexes	**78**	**39**

* dna-test01 and dna-dev01 are new plexes to be provisioned. Dev/test node spec: 2 vCPU / 7.9 GB. Prod node spec: 4 vCPU / 15.8 GB. Medium Node equivalents calculated as vCPUs ÷ 2.

5 Workload Analysis & PS Review Candidates (90-day)

Pipeline execution patterns by Snaplex. Data covers Jan 6 – Mar 26 2026. Candidates identified based on high volume, long duration, and failure rate.

Pipeline Execution Summary by Snaplex

SNAPLEX	INVOKER TYPE	STATE	TOTAL EXECUTIONS	TOTAL DOCS	AVG DURATION	ERROR RATE	PS FLAG
`prod01`	nested_pipeline	Completed	**9,571,000**	113 B	**111 hrs avg**	2.8% err docs	CRITICAL
`prod01`	scheduled	Completed	1,064,000	3.2 B	**60 hrs avg**	0.24%	CRITICAL
`prod01`	nested_pipeline	Failed	133,000	4.9 B	8.4 hrs avg	**1.37% of nested**	HIGH
`prod01`	error_handler	Completed	222,905	2.6 M	2.4 hrs avg	—	HIGH
`prod01`	triggered	Completed	178,432	110 M	1.6 hrs avg	<0.1%	REVIEW
`prod01`	scheduled	Stopped	67	64 M	**13 hrs avg**	—	REVIEW
`prod02 (DnA)`	nested_pipeline	Completed	11,424	14.4 M	2.2 hrs avg	6.9% err docs	WATCH
`dev01`	nested_pipeline	**Failed**	**603,220**	224 M	4.3 hrs avg	**86% fail rate**	CRITICAL
`dev01`	scheduled	Completed	63,259	136 M	4.4 hrs avg	2.9%	REVIEW
`autosync`	nested_pipeline	Completed	154	10,780	10 min avg	23% err rate	IDLE

Priority PS Candidates

1. **Prod01 — Long-running nested & scheduled pipelines (avg 60–111 hrs)**
9.57M nested pipeline executions with an average runtime of *111 hours* are the primary driver of prod01's CPU saturation. These pipelines need architectural review: are they truly long-running streaming jobs, or batch loops that could be parallelised, chunked, or scheduled at off-peak times? Professional Services should audit the top 10 most CPU-consuming pipelines directly in the platform.

2. **Dev01 — 86% nested pipeline failure rate (603K failures vs 101K completions)**
This is not normal dev churn — 603,000 failures over 90 days at 4.3 hours average duration means hundreds of thousands of hours of compute consumed by failing pipelines. This strongly suggests pipelines being tested/retried against broken logic or data, inadequate error handling, or integration issues. PS review of dev pipeline lifecycle and error handling strategy is urgently needed.

3. **Prod01 — 222K error_handler executions (2.4 hrs average)**
The volume of error handler completions suggests widespread pipeline errors being caught but processed. This is expensive: each error handling run averages 2.4 hours. PS should review whether error handling logic can be simplified, or whether root cause fixes would eliminate this overhead entirely.

4. **Prod02 (DnAGroundplex) — 6.9% error doc rate on early-stage pipelines**
The new DnA plex has elevated error document rates on its nested pipelines (6.9% vs <3% on prod01). As this is early-stage, PS engagement now could prevent these patterns becoming embedded. Investigate data quality or pipeline logic before the load scales up.

🔍 How to Identify the Specific Pipelines for PS Review

The BigQuery analytics layer aggregates execution data at *snaplex + invoker type* level — individual pipeline names are not exposed in the standard analytics tables. To name the specific pipelines, PS must use the SnapLogic platform directly. Here is exactly what to do:

A

Prod01 — Long-running nested pipelines (target: 111 hrs avg, 9.57M execs)
In SnapLogic Platform Manager → Org: `SyngentaProd` → Runtime → Pipeline Executions.
Filter: *Snaplex = prod01, Invoker = nested_pipeline, State = Completed*. Sort by **Duration descending**. The top 10–15 pipelines by cumulative runtime are the PS review targets — look for patterns running >24 hrs per execution, high repeat frequency, or large document volume.
Likely candidates: batch ETL loops, large-dataset transforms, pipelines processing flat files record-by-record rather than in bulk.

B

Prod01 — Scheduled pipelines running 60 hrs avg (1.06M execs)
Filter: *Snaplex = prod01, Invoker = scheduled, State = Completed*. Sort by **Duration descending**. Any scheduled pipeline with avg runtime >4 hours is a candidate for chunking or parallelisation review.
A 60-hr average on a scheduled pipeline strongly suggests it is processing a large dataset in serial. PS should review whether bulk-mode snaps or parallel child pipelines can reduce this by 10–20×.

C

Dev01 — 86% failure rate on nested pipelines (603K failures)
Filter: *Snaplex = dev01, Invoker = nested_pipeline, State = Failed*. Sort by **Execution count descending**. The top 5 pipeline paths by failure volume are the immediate PS targets — these are consuming hundreds of thousands of hours of dev compute with no output.
At 86% fail rate across 603K runs, this is almost certainly a small number of pipelines running in an automated retry loop or being repeatedly triggered against broken data/logic. One or two root-cause fixes likely resolve the majority.

D

Prod01 — 222K error_handler executions (2.4 hrs avg)
Filter: *Snaplex = prod01, Invoker = error_handler, State = Completed*. These are not separate pipelines but error handling sub-pipelines triggered by parent failures. To identify the parents: look at pipelines with the highest error_handler trigger rate via Platform Manager → Pipeline Overview → Error Handling panel.
The parent pipelines generating 222K error handlers are likely the same nested/scheduled pipelines above. PS should confirm whether error handling is being used as a retry mechanism (expensive) vs genuine exception capture.

Note on BigQuery data availability: SnapLogic's analytics warehouse (codedugong) does not store individual pipeline names in the standard execution tables — data is aggregated at the org/snaplex/invoker_type level. The patterns above are derived from aggregate telemetry. Individual pipeline identification requires direct access to the SnapLogic Platform Manager for `SyngentaProd` and `syngenta_dev` orgs.

6 Summary of Recommendations

1. **PS engagement on prod01 long-running pipelines** — 9.57M executions averaging 111 hours is the root cause of CPU saturation. PS should audit top-volume pipelines for parallelism, chunking, and scheduling improvements, and review the 133K failure rate.

2. **PS engagement on syngenta_dev failure patterns** — 86% nested pipeline failure rate in dev (603K failures over 90d) indicates systemic pipeline quality issues. PS should review dev lifecycle management, error handling strategy, and retry patterns.

3. **Provision DnA Groundplex mirrors in Test and Dev (2 nodes each)** — the new prod02 DnAGroundplex has no CI/CD path. DnA pipelines cannot be validated before production promotion until Test and Dev mirrors exist. This is the primary CI/CD gap to close.

4. **Expand SyngentaTest Groundplex (+2 nodes)** — currently 3 nodes (6 vCPUs) vs prod's 9 nodes (36 vCPUs). Insufficient for CI/CD gating of prod-scale workloads. Add 2 nodes after reviewing the 1 idle node currently present.

5. **Monitor DnAGroundplex (prod02) and prod01 over next 60 days** — prod02 has 4 nodes provisioned and is at 2% CPU as DnA workloads onboard. As pipelines migrate from prod01 to prod02, prod01 CPU pressure will reduce. If prod01 remains above 60% after DnA migration completes, reassess +3 node addition then.